Year 10 Higher

REVISION PACK

Number 2

Name: _____

Class: _____

Number 2 · Percentages (Lesson 4.1)

Worked example: percentage loss

The table shows what Sara paid for a car and what Sara sold it for. Work out Sara's percentage loss.

Bought for	Sold for
KD 2500	KD 2100

Step 1	Loss	$2500 - 2100 = 400$
Step 2	Loss ÷ cost × 100	$\frac{400}{2500} \times 100 = 16\%$
Step 3	Answer	16%

Practice: percentage loss

1.1 The table shows what Ali paid for a car and what Ali sold it for. Work out Ali's percentage loss.

Bought for	Sold for
KD 18 000	KD 14 940

1.2 The table shows what Mona paid for a sofa and what Mona sold it for. Work out Mona's percentage loss.

Bought for	Sold for
KD 640	KD 528

1.3 The table shows what Omar paid for a boat and what Omar sold it for. Work out Omar's percentage loss.

Bought for	Sold for
KD 2500	KD 2150

1.4 The table shows what Huda paid for a car and what Huda sold it for. Work out Huda's percentage loss.

Bought for	Sold for
KD 12 000	KD 9000

1.5 The table shows what Fahad paid for a TV and what Fahad sold it for. Work out Fahad's percentage loss.

Bought for	Sold for
KD 450	KD 396

1.6 The table shows what Laila paid for a car and what Laila sold it for. Work out Laila's percentage loss.

Bought for	Sold for
KD 8000	KD 7400

Number 2 · Percentages (Lesson 4.1)

Worked example: reverse percentages

The value of a car decreases by 15% each year. After one year it is worth KD 5100. Work out the value of the car when new.

Step 1	After one year	$100\% - 15\% = 85\%$
Step 2	Multiplier: 85% = 0.85	0.85
Step 3	Divide by the multiplier	$5100 \div 0.85 = 6000$
Step 4	Answer	$\text{KD } 6000$

Practice: reverse percentages

1.7 The value of a car decreases by 24% each year. After one year it is worth KD 5320. Work out the value of a car when new.

1.8 The value of a motorbike decreases by 15% each year. After one year it is worth KD 3060. Work out the value of a motorbike when new.

1.9 The value of a laptop decreases by 8% each year. After one year it is worth KD 391. Work out the value of a laptop when new.

1.10 The value of a phone decreases by 15% each year. After one year it is worth KD 204. Work out the value of a phone when new.

1.11 The value of a boat decreases by 5% each year. After one year it is worth KD 2850. Work out the value of a boat when new.

1.12 The value of a van decreases by 20% each year. After one year it is worth KD 8320. Work out the value of a van when new.

Number 2 · Recurring decimals (Lesson 4.2)

Worked example: recurring decimal proofs

Prove algebraically that $0.1\dot{6} = \frac{1}{6}$

Step 1	Let x be the decimal	$x = 0.1666\ldots$
Step 2	Multiply by 100	$100x = 16.666\ldots$
Step 3	Multiply by 10	$10x = 1.666\ldots$
Step 4	Subtract the two equations	$90x = 15$
Step 5	Divide by 90, then simplify	$x = \frac{15}{90} = \frac{1}{6}$

Practice: recurring decimal proofs

2.1 Prove algebraically that $0.2\dot{7} = \frac{5}{18}$

2.2 Prove algebraically that $0.3\dot{8} = \frac{7}{18}$

2.3 Prove algebraically that $0.4\dot{1} = \frac{37}{90}$

2.4 Prove algebraically that $0.1\dot{5} = \frac{7}{45}$

2.5 Prove algebraically that $0.2\dot{5} = \frac{23}{90}$

2.6 Prove algebraically that $0.5\dot{2} = \frac{47}{90}$

Number 2 · Percentages (Lesson 4.1)

Worked example: increasing and decreasing by a percentage

A laptop costs KD 650. In a sale, the price is reduced by 18%. Work out the sale price.

Step 1	Find the multiplier	$100\% - 18\% = 82\% = 0.82$
Step 2	Amount × multiplier	$0.82 \times 650 = 533$
Step 3	Answer	KD 533

Practice: increasing and decreasing by a percentage

3.1 A coat costs KD 480. In a sale, the price is reduced by 15%. Work out the sale price.

3.2 A sofa costs KD 1250. In a sale, the price is reduced by 22%. Work out the sale price.

3.3 A pair of trainers costs KD 86. In a sale, the price is reduced by 35%. Work out the sale price.

3.4 A car loses 7.5% of its value in one year. At the start of the year it was worth KD 2400. Work out its value at the end of the year.

3.5 A phone costs KD 340. The price increases by 18%. Work out the new price.

3.6 Rashid earns KD 15 000 a year. His salary increases by 3.2%. Work out his new salary.

Number 2 · Percentages (Lesson 4.1)

Worked example: percentage increase

Sam's salary increased from KD 24 000 to KD 25 080. Work out the percentage increase.

Step 1	Find the increase	$25080 - 24000 = 1080$
Step 2	Increase ÷ original × 100	$\frac{1080}{24000} \times 100 = 4.5$
Step 3	Answer	4.5%

Practice: percentage increase

3.7 Huda's salary increased from KD 32 000 to KD 33 440. Work out the percentage increase.

3.8 The price of a watch increased from KD 250 to KD 285. Work out the percentage increase.

3.9 The rent of a flat increased from KD 1800 to KD 1953. Work out the percentage increase.

3.10 The number of students in a club increased from 64 to 80. Work out the percentage increase.

3.11 The population of a town increased from 12 500 to 13 250. Work out the percentage increase.

3.12 The price of a ticket increased from KD 75 to KD 93. Work out the percentage increase.